

Partitives, groups, and wholes

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There's an attractive view in social ontology on which a group is identical to its members. It has its defenders, but seems felled by a simple objection resting on plausible assumptions about the semantics of plural partitives—assumptions built into standard plural logic. We argue the assumptions are false: whether something counts as 'one of' a plurality turns on how that plurality is carved up, a sensitivity the standard regimentation misses. We propose and defend a cover-based semantics that captures it, blocks the objection, and earns its keep on independent grounds. The same account answers a parallel, well-known objection to composition as identity. Our view thus bears both on the logic and meaning of plurals and on debates in metaphysics.

Keywords: composition as identity; cover; group; numeral; partitive; plural logic.

I. A problem for the identity theory of groups

In recent years, a great deal of attention has been devoted to the metaphysics of social groups such as teams, boards, courts, juries, and governments. On many views, a group is an individual, distinct from the individuals making it up: a pair of people, for instance, is an individual that has two people as members.¹ On an opposing view, a group just *is* the individuals making it up: a pair *is* two people; a team *is* its members; and so on.² Call this the *identity theory of groups*. According to the theory, collective nouns like 'team' are *pseudo-singular* (Oliver & Smiley 2016: 306): while they behave grammatically like singular

¹See, for instance, Barker (1992), Schwarzschild (1996, ch. 9), Uzquiano (2004b), Ritchie (2013), and Snyder & Shapiro (2021).

²See notably Magri (2012), Korman (2015), and Horden & López de Sa (2021).

nouns, they are semantically plural; a collective noun is one that is true of *some things* rather than a single thing—a *plurality* rather than an individual.

Outside the field of social ontology, the idea that at least *some* collective nouns are pseudo-singular is quite popular. It seems intuitive that, for instance, a pair of socks just is the two socks that make it up; ‘pair of socks’ is a noun that is true of *two individuals* rather than any one individual.³ Those who accept this are sometimes reticent to extend it to *all* collective nouns,⁴ but the identity theorist insists that what goes for ‘pair of socks’ goes for ‘team’, ‘board’, ‘court’, ‘jury’, and the like.

We’re sympathetic to the identity theory, but unfortunately, it appears to be undermined by the meaning and logic of plural partitives. Plural partitives are expressions of the form [Det [of NP]] with the embedded noun phrase plural. Some examples are ‘one of the guests’, ‘two of several customers who complained’, and ‘some of them’. (We focus on cases where the noun phrase is definite.) According to Ionin & Matushansky (2018) and Marty (2019), the semantic function of the preposition ‘of’ is to indicate that any individuals satisfying the entire partitive expression must be among the individuals denoted by the definite plural noun phrase. The determiner then adds further information—e.g. that the satisfiers are *n* in number. This simple and plausible analysis is what creates the problem for the identity theory.

Consider:

- (1) One of the players is Québécois.
- (2) One of the teams is Québécois.

Suppose the teams are Ann and Bill, on one hand, and Carrie and Daniel, on the other. According to the identity theory, both ‘the teams’ and ‘the players’ denote the four people, and so the analysis of partitives predicts that both (1) and (2) are true just in case one player is Québécois. Now suppose (i) that Daniel is from Québec, while the other three players are from other provinces in Canada and (ii) that both teams are affiliated with cities in the province of Ontario. Then, we have one Québécois player but no Québécois teams. So, (1) is true, while (2) is false. How can that be, if the players *are* the teams?⁵

The argument can apparently be bolstered using the resources of *plural logic*. We assume a *one-sorted* plural language, in which all terms are *neutral*: a constant (*‘a’, ‘b’, ...*) may denote either one individual or many, and likewise a variable (*‘x’, ‘y’, ...*) may stand in for either one individual or many.⁶ The language has a single primitive logical predicate, *inclusion*, ‘ $\sqsubset$ ’, read ‘*a* is/are

³See e.g. Black (1971), Simons (1982b, 1987: 144–8), Oliver & Smiley (2016: 138–40, 303–7; 2022), and Horden & López de Sa (2021).

⁴See e.g. Simons (1982b: 208–11, 1987: 144–8).

⁵See Uzquiano (2004b: 144–5) for a similar argument.

⁶One-sorted languages stand in contrast to two-sorted languages, which contain *singular* and *plural* terms as distinct syntactic categories. See Rayo (2002: 451–3).

among b . ' $a \sqsubseteq b$ ' is true just in case every individual denoted by ' a ' is denoted by ' b ' (though not necessarily vice versa). Inclusion is reflexive and transitive.

The language contains three defined logical predicates. The first is identity (' $=$ '):

Identity $a = b \equiv_{\text{df}} a \sqsubseteq b \ \& \ b \sqsubseteq a$

a and b are identical just in case ' a ' and ' b ' denote all and only the same things. The second is an *individuality* predicate, ' I ':

Individuality $Ia \equiv_{\text{df}} \forall x(x \sqsubseteq a \rightarrow x = a)$

An individual is the limit case of a plurality: it's a plurality that includes only one thing, namely itself. Finally, there's a *singular inclusion* predicate, ' $\leq$ ':

Singular inclusion $a \leq b \equiv_{\text{df}} Ia \ \& \ a \sqsubseteq b$

a is singularly included in b just in case a is an *individual* included in b . A typical plural logician reads ' $a \leq b$ ' as ' a is one of b '.

With these resources in place, the plural logician regiments our sentences (1) and (2) as follows:

(1*) $\exists x(x \leq p \ \& \ \text{QUÉBÉCOIS}(x))$

(2*) $\exists x(x \leq t \ \& \ \text{QUÉBÉCOIS}(x))$

where ' p ' and ' t ' denote the players and the teams, respectively. Crucially, ' $x \leq y$ ' is treated as a *transparent* context: whether $x \leq y$ holds turns entirely on what ' x ' and ' y ' denote. So if ' p ' and ' t ' co-denote, then $x \leq p$ just in case $x \leq t$, and (1*) and (2*) are equivalent. If the regimentation is correct, then the teams can be identical to the players only if (1) and (2) are equivalent. Since (1) and (2) are *not* equivalent, she concludes that the teams aren't identical to the players. A seemingly attractive view of the metaphysics of groups has apparently been felled by plural logic.

It may be asked why a defender of a *metaphysical* thesis should care whether some *semantics* vindicates her view. The answer is that the objection is itself a piece of reasoning conducted in English: it's stated in English, and whether it's valid or sound therefore turns on the semantics of English. Regimentation into plural logic can obscure this dependence since ' $\leq$ ' is presented as a transparent logical primitive about which there is nothing further to decide. But the regimentation earns its force only on the assumption that ' $x \leq y$ ' correctly translates the English ' x is one of y '. That assumption isn't a logical truth; it's a substantive semantic claim. If it fails, the regimentation fails, and the objection lapses with it. So the semantics isn't an idle addendum to the metaphysics: it's the very thing on which the cogency of the objection depends. Our task is accordingly to examine that assumption directly. We'll argue that it ought to be rejected.

The identity theory of groups, however, is only one instance of a more general metaphysical idea: that a complex object just *is* the things that make it up. Its best-known instance is *composition as identity*, the view that a whole is identical to its parts. As we'll show in Section V, that view faces an objection exactly parallel to the one just rehearsed, turning on the very same partitive construction and the same transparent reading of 'is one of'. Our strategy is therefore uniform. We first diagnose the flawed assumption about partitives (Section II). We then set out our own formal framework (Section III) and use it to give a semantics for partitive expressions like 'one of the teams' that defuses the objection to the identity theory of groups (Section IV). Finally, we show that this same semantics defuses the parallel objection to composition as identity (Section V).

II. A sketch of a solution

Elsewhere (Nicolas & Payton 2025b), we've argued that plural predications in general must be understood in terms of *generalized covers*. Informally, a cover is a way of carving a plurality into possibly overlapping subpluralities, without remainder. For example, since a pair of socks just is the two socks making it up, two pairs of socks just are the four socks making them up. We can carve those four socks into four individual socks or into two pairs:

$$\begin{array}{rcl}
 & \longrightarrow & s_1 \\
 & \longrightarrow & s_2 \\
 s_1 s_2 s_3 s_4 & \longrightarrow & s_3 \\
 & \longrightarrow & s_4 \\
 & \longrightarrow & s_1 s_2 \\
 & \longrightarrow & s_3 s_4
 \end{array}$$

The intuitive idea behind our approach is that the application-conditions of plural predicates can be sensitive to such carvings: whether a predicate is true of some things can depend not merely on what those things are, but on how they have been carved up.

This idea has independent motivation. Consider this example, inspired by one of Linnebo & Nicolas (2008: 193):

(3) The red socks, the wool socks, and the perforated socks overlap.

Intuitively, (3) is true just in case the three pluralities of socks have a member in common, i.e. just in case some sock is red, wool, and perforated. But if a plurality of socks just is the socks making it up, then the subject term refers to a single plurality of socks, and the division of those socks into three groups—on which the truth-conditions turn—is lost.

The standard remedy posits *higher-level plurals*: the idea is that, just as we can plurally refer to and quantify over *individuals*, we can plurally refer to and quantify over *pluralities* of individuals.⁷ On this view, the subject in (3) denotes not the individual socks but a plurality of three sock-pluralities, and the sentence says that *they* overlap. The coherence of higher-level plural reference is, however, contested,⁸ and even granting that the idea is coherent, it threatens to bring with it unwanted ontological commitments, forcing us to distinguish the sock-pluralities from the socks making them up (Payton 2025).

Covers offer an alternative with no such commitments. The subject denotes one plurality of socks, and ‘overlap’ is true of it relative to a carving into the red, the wool, and the perforated socks—three possibly overlapping subpluralities—just in case some sock falls in all three.⁹ No higher-level plural reference is needed; the work previously done by a plurality of pluralities is now done instead by a single plurality *under* a cover. Note that everything we’ve said here is independent of the metaphysical views of groups and wholes with which we’re concerned in this paper. A theorist with no stake in those debates, but who doubts the coherence of higher-level plural reference, has good reason to adopt covers.

We can now informally state our solution to the problem of Section I. The plural logician regimentes the English ‘is one of’ as ‘ $\leq$ ’, and treats ‘ $a \leq b$ ’ as *transparent*: its truth turns entirely on what ‘ a ’ and ‘ b ’ denote—true if a is an individual among the denotation of ‘ b ’, false otherwise. Thus, on her approach, if the players are identical to the teams, then ‘one of the players’ and ‘one of the teams’ denote *the same property*, the property of being included in that plurality that is denoted by both ‘the players’ and ‘the teams’. Hence the demand that, on the identity theory, (1) and (2) come out equivalent.

On our view, the plural logician’s regimentation of ‘is one of’ is mistaken in a specific way, traceable to the numeral. A partitive ‘one of the F s’ makes two demands on its satisfiers, contributed by two different words:

- (i) The satisfiers must, as the plural logician says, be *among* the F s. This condition is contributed by ‘of’, and is cover-insensitive: it depends only on what ‘the F s’ denotes.
- (ii) The satisfiers must constitute exactly *one cell* of a special cover of the F s. This is what we call the *basic cover* $\star^F$ of the head noun ‘ F ’: the carving that divides any suitable plurality into all and only the F s within it. (We define basic covers more precisely in Section III.3.) This condition is contributed

⁷See e.g. Rayo (2006), Oliver & Smiley (2016, ch. 15), Simons (2016), Florio & Linnebo (2021, ch. 9), and Grimaud (2021a).

⁸See especially Simons (1982a: 187–95), Uzquiano (2004a: 438–40), McKay (2006: 46–53, 137–9), and Ben-Yami (2013).

⁹For more discussion of this example and other alleged cases of higher-level plural predication, see Nicolas & Payton (2025b).

by the numeral, ‘one’, and is *cover-sensitive*: it depends not only on what ‘the *F*s’ denotes but on how the basic cover $\star^F$ carves those things up.

Now, the head nouns in (1) and (2) contribute different basic covers to the partitives in which they occur: ‘the players’ contributes $\star^P$, which carves its denotation into individual players, while ‘the teams’ contributes $\star^T$, which carves the very same denotation into teams. So, due to the cover-sensitivity of partitives, ‘one of the players’ and ‘one of the teams’ denote different properties, *even if ‘the players’ and ‘the teams’ are co-denoting*. This means that (1) and (2) are *not* equivalent: (1) requires at least one Québécois player, whereas (2) requires at least one Québécois team, and so (1) can be true while (2) is false.

In what follows, we make this solution precise. We present a formal framework yielding both an *analysis* of partitive sentences—a statement of their truth-conditions—and a *compositional semantics*—an account of how those conditions arise from the meanings of those sentences’ parts. Partitives like ‘one of’ are shown to be cover-sensitive in the way just described, blocking the argument against the identity theory.

III. A framework

III.1 A formal language

The formal language in which we’ll provide our analyses is a modification of the first-order plural language CPL, developed in Payton (2025).

The foundation of this language is similar to the one-sorted plural language sketched in Section I: it contains neutral constants and variables, a basic inclusion predicate, and defined identity, individuality, and singular inclusion predicates.¹⁰ Onto this foundation, we add generalized covers, denoted by the Latin letters ‘*i*’, ‘*j*’, ‘*k*’, etc. As we said, a cover can be thought of informally as a way of carving a plurality into subpluralities. More formally, a cover *i* is a partial, multivalued function; when *i* applies to a plurality *a*, its values (or cells) are subpluralities of *a*.¹¹ We define these functions as follows:

Covers *i* covers *a* just in case:

- (i) Every value *i* assigns to *a* is included in *a*.
- (ii) Every individual included in *a* is included in some value *i* assigns to *a*.

¹⁰Payton originally develops CPL as a *two-sorted* language (see footnote 6). This is why the language we use here is a *modification* of CPL.

¹¹This notion of a generalized cover is developed more fully in Nicolas & Payton (2025b). Note that instead of functions, one–many relations between pluralities and subpluralities can be used; for our purposes, nothing hinges on the choice.

When $i(a)$ is defined, it either divides a into subpluralities, or it ‘does nothing’, returning a themselves. Of course, if a is an individual, then the only thing included in a is itself, and so a cover can only map a to itself.

In CPL, a term can only occur as an argument of a predicate if it’s indexed to a cover: ‘ Fa ’ isn’t a formula, but ‘ Fa^i ’ is. Likewise for variables: ‘ $\exists x(Fx)$ ’ isn’t a formula, but ‘ $\exists x(Fx^i)$ ’ is.

The application-conditions of a predicate are allowed to be sensitive, not merely to the denotations of the terms which fill its argument-places, but also to the covers to which those terms are indexed. So, whether ‘ F ’ is true of a can depend on our choice of cover. However, predicates aren’t *required* to be cover-sensitive in this way. The logical predicate ‘ $\sqsubseteq$ ’, for instance, is cover-insensitive: ‘ $a^i \sqsubseteq b^j$ ’ is true for all choices of i and j if it’s true for any of them. Likewise for all logical predicates defined using ‘ $\sqsubseteq$ ’.

CPL contains a new, defined predicate that will be useful for our purposes. This is a cover-sensitive analogue of inclusion, c-inclusion ($\lesssim$), defined in the meta-language as follows:

c-inclusion ‘ $a^i \lesssim b^j$ ’ is true iff a are a value j assigns to b .

The claim that a are c-included in b , with respect to i and j respectively, is an object-language correlate of the meta-language claim that a are a value of $j(b)$, and so its truth depends on our choice of cover. (Note that ‘ $\lesssim$ ’ is only cover-sensitive at its second argument-place: whether a are c-included in b depends on how b are carved up, but not on how a are.) Given the definition of cover, c-inclusion implies neutral inclusion.

c-inclusion is used to define the distributive variants of predicates in CPL:

Distributive predicates $F^D(a^i) \equiv_{\text{df}} \exists x(x^i \lesssim a^i) \ \& \ \forall x(x^i \lesssim a^i \rightarrow F(x^i))$

‘ F^D ’ is true of a with respect to i just in case i is defined at a and F is true of every value i assigns to a . So, whether ‘ F^D ’ is true of a depends on our choice of cover.¹²

III.2 Compositional semantics

Our approach to compositional semantics is a modification of the familiar type-theoretic approach exemplified by Heim & Kratzer (1998).¹³ Our basic semantic types are pluralities (type e), the truth-values 1 and 0 (type t), and covers (type c). Other semantic types are constructed recursively from these; for instance, the extensions of one-place verb phrases are functions from pluralities to truth-values (type <et>).

¹²The first conjunct matters: without it, ‘ F^D ’ would be vacuously true wherever i is undefined. Thanks to Annica Vieser for discussion.

¹³The approach is developed more fully in Nicolas & Payton (2025a).

The semantics is *intensional*: the denotation of an item in the lexicon is an *intension*, i.e. a function from one or more indices to some entity in the type-theoretic hierarchy. For our purposes, the only indices we'll need are covers. ' $\llbracket \alpha \rrbracket$ ' refers to the denotation of the lexical item α , i.e. an intension that hasn't yet been saturated by any covers. We use superscripts to indicate that one or more argument-places for covers have been saturated: $\llbracket \alpha \rrbracket^i$ is the result of saturating an argument-place with the cover i , $\llbracket \alpha \rrbracket^{i,j}$ is the result of saturating one argument-place with the cover i and another with the cover j , etc.

For our purposes, we'll need only two compositional rules. The first is a variant of Functional Application (Heim & Kratzer 1998: 44):

Functional Application (FA) If α is a branching node and β and γ are its daughters, and $\llbracket \beta \rrbracket^i$ is a function whose domain includes $\llbracket \gamma \rrbracket^j$, then $\llbracket \alpha \rrbracket^{i,j} = \llbracket \beta \rrbracket^i(\llbracket \gamma \rrbracket^j)$.

The second is a variant of Intensional Functional Application (Heim & Kratzer 1998: 308):

Intensional Functional Application (IFA) If α is a branching node, and β and γ are its daughters, then for any i , if $\llbracket \beta \rrbracket^i$ is a function whose domain includes $\lambda j(\llbracket \gamma \rrbracket^{j,k})$, then $\llbracket \alpha \rrbracket^{i,k} = \llbracket \beta \rrbracket^i(\lambda j(\llbracket \gamma \rrbracket^{j,k}))$.

IFA allows an item in the lexicon to operate, not only on the extensions of other items, but on their intensions.

III.3 Basic and fitting covers

As we said above, and have argued elsewhere in detail, the truth or falsity of a sentence can depend on how the denotation of the subject term gets carved up. It's worth stressing that this motivation is wholly independent of the metaphysical disputes at issue here. The point is a familiar one in the semantics of distributivity. Consider:¹⁴

(4) Rodgers, Hart, and Hammerstein wrote musicals.

This sentence is true on the carving that groups the three men into Rodgers-and-Hart and Rodgers-and-Hammerstein—saying of each *pair* that they wrote musicals—but false on the carving into the three individuals, none of whom wrote musicals alone. Such readings are recognized in natural language semantics quite independently of any view about groups or wholes, and they already require something like covers. Covers in the form we adopt are thus not a device tailored to the present problem; they earn their keep elsewhere.

¹⁴Inspired by example (30) of Gillon (1987).

However, there appear to be limits on how pluralities can be carved up in the interpretation of natural language sentences; not just *any* cover is available for the interpretation of a given sentence. Compare:

- (5) The people lifted the piano.
- (6) The pairs (of people) lifted the piano.

There's a reading of (5) on which 'lifted the piano' distributes over each person and the sentence is true just in case each *person* lifted the piano. There's also a reading of (5) on which the predicate distributes over pairs of people and the sentence is true just in case each *pair* lifted the piano. But while (6) admits the latter reading, it doesn't admit the first; there's no reading of (6) on which 'lifted the piano' distributes over people rather than pairs.

On our view, this is because count nouns and plural definites are associated with special covers, and predicates are semantically constrained to be interpreted with respect to these. First, there are *fitting* covers:

Fitting covers A cover i fits ' F ' if it satisfies these two conditions:

- (i) For any x , $i(x)$ is defined whenever there is some cover j of x such that: $F^D(x^j)$.
- (ii) When $i(x)$ is defined, its values are such that: $\forall y(y^i \lesssim x^i \rightarrow F^D(y^i))$.

That is, a cover fits ' F ' just in case, when defined for some x , it carves x into F s and/or arbitrary combinations thereof. Second, there are *basic* covers:

Basic covers The basic cover $\star^F$ is the cover that fits ' F ' and whose values, whenever it's defined for some x , satisfy: $\forall y(y^{\star^F} \lesssim x^{\star^F} \leftrightarrow (y^{\star^F} \sqsubseteq x^{\star^F} \ \& \ F(y^{\star^F})))$.¹⁵

That is, the basic cover of ' F ', $\star^F$, is the fitting cover that, when defined for some x , carves x into all and only the F s included in x . (No F s may be left out, and arbitrary combinations of F s aren't allowed.)

Nominal expressions headed by a noun inherit the fitting and basic covers of that noun. For instance, 'person', 'people', and 'the people' all share the basic cover $\star^{\text{Per}}$, which, when defined for some x , carves them into individual people. Likewise, 'pair of people', 'pairs of people', and 'the pairs of people' all share the basic cover $\star^{\text{Pair}}$, which, when defined for some x , carves them into pairs of people.¹⁶ Inheritance extends to bound variables: in a quantified sentence,

¹⁵We should record a difference in notation with Nicolas & Payton (2025a: 15). Where that paper used the ' $\leq$ ' symbol for inclusion, we use the symbol ' $\sqsubseteq$ ' for that purpose; and we reserve the symbol ' $\leq$ ' for *singular inclusion*.

¹⁶The derivations which follow treat collective nouns as *cover-insensitive*: whether some things are a team, or a pair of people, doesn't vary with the cover at which the noun is evaluated. In Nicolas & Payton (2025a), by contrast, 'pair of people' is given a cover-sensitive entry: some people are a pair with respect to i just in case i carves them into two people. While nothing

when such a nominal expression restricts a quantified variable, that variable inherits its basic and fitting covers.¹⁷

While predicates in CPL can be cover-sensitive, our choice of cover is constrained to those which *fit* the predicate associated with the head noun. This explains the different readings of (5) and (6). In (5), ‘lifted the piano’ must be interpreted with respect to a cover which fits ‘the people’: so, it can be read as distributing over individual people or over combinations of people (e.g. *pairs* of people). But in (6), ‘lifted the piano’ must be interpreted with respect to a cover which fits ‘the pairs’: so, it can be read as distributing over pairs of people, but *not* as distributing over individual people.¹⁸

Note that the denotation of the predicate ‘lifted the piano’ doesn’t vary with our choice of cover: rather, the same denotation is offered a different cover as argument. This is crucial, since otherwise a sentence like

(7) On Monday the people lifted the piano, and on Tuesday the pairs did so.

would pose a serious problem. (7) sure looks like an instance of VP-deletion, with ‘did so’ getting its denotation from the antecedent, ‘lifted the piano’. If instead it needed a different denotation, there would be a puzzle as to how that denotation had been assigned. On our view, no such puzzle arises: ‘did so’ is assigned the exact same denotation as ‘lifted the piano’, just as we’d expect. The different reading results from the fact that the subject ‘the pairs’ contributes a different cover as argument than ‘the people’.

IV. The problem solved

With this framework in place, we can now give our account of the meanings of partitive expressions—and the solution to the problem for the identity theory of groups that depends on it—in more detail.

Recall our sentences:

- (1) One of the players is Québécois.
- (2) One of the teams is Québécois.

hinges on this difference in the present paper, the cover-sensitive approach runs into trouble that the cover-insensitive approach avoids; see Nicolas & Payton (n.d.) *Mass Nouns and Plural Logic Revisited*.

¹⁷This is the cover-theoretic form of the conservativity of natural-language determiners. Conservativity is the property that $\text{det}(F)(G)$ holds iff $\text{det}(F)(F \cap G)$ holds, so that the nuclear scope G is evaluated only over satisfiers of the restrictor F (Barwise & Cooper 1981). Our constraint strengthens this: the restrictor fixes not only *which* things the scope is evaluated on, but *how* they are carved—via its basic cover—so that distribution in the scope bottoms out at restrictor-sized cells.

¹⁸See Nicolas & Payton (2025a) for more details.

In these sentences, a numerically specific plural partitive occupies subject position. Following a common thought, we assume that numeral phrases in subject position aren't the main source of quantificational force, and that this is introduced by way of an unpronounced existential operator:

$$[\exists\exists] = \lambda F_{\langle et \rangle} \lambda G_{\langle et \rangle} [\exists x(F(x) \& G(x))]$$

This operator takes two functions of type $\langle et \rangle$ to the proposition that both functions are true of some x . Note that the bound occurrences of ' x ', here, aren't indexed to covers. That's because the indexing will be achieved compositionally. Specifically, ' x ' is restricted by the partitive, and so inherits its basic and fitting covers (Section III.3).

Starting with (2), we obtain¹⁹

$$\begin{aligned} & [\text{One of the teams is Québécois}]^{i,j} \\ &= [\exists\exists](\llbracket \text{one of the teams} \rrbracket^i)(\llbracket (\text{is}) \text{ Québécois} \rrbracket^j) \quad \text{By FA} \\ &= \exists x(\llbracket \text{one of the teams} \rrbracket^i(x) \& \llbracket (\text{is}) \text{ Québécois} \rrbracket^j(x)) \end{aligned}$$

That is, some x are both (i) one of the teams and (ii) Québécois. The meaning of $\llbracket (\text{is}) \text{ Québécois} \rrbracket^j$ is easily explicated:

$$\llbracket (\text{is}) \text{ Québécois} \rrbracket = \lambda i \lambda x [\text{QUÉBÉCOIS}(x^i)]$$

The meaning of $\llbracket \text{one of the teams} \rrbracket^i$, however, must itself be composed:

$$\llbracket \text{one of the teams} \rrbracket^i = \llbracket \text{one} \rrbracket^i (\llbracket \text{of the teams} \rrbracket) \quad \text{By IFA}$$

In giving the derivation, our task will be to discharge the two conditions identified informally in Section II: (i) being included in the relevant plurality, which is contributed by 'of'; and (ii) being exactly one cell of the head noun's basic cover, which is contributed by 'one'.

We give the following semantic values to 'of' and 'the teams':

$$\llbracket \text{of} \rrbracket = \lambda i \lambda x \lambda y [y^i \sqsubseteq x^i]$$

$$\llbracket \text{the teams} \rrbracket = \lambda i [t]$$

Each of these deserves some comment.

First, we're treating 'the teams' as denoting a simple function from a cover to a plurality (namely, the teams). The actual semantics of plural definites like 'the teams' is more complex. On our view, for any count noun ' F ', 'the F s' denotes the largest (contextually salient) plurality of F s; that is, it denotes the x that can be carved into all and only the (contextually salient) F s that there are.²⁰

¹⁹We annotate the steps licensed by our two composition rules: 'By FA' abbreviates 'By Functional Application' and 'By IFA' abbreviates 'By Intensional Functional Application'. Steps of β -conversion and the like are left unmarked.

²⁰More formally, 'the F s' denotes the unique x , if any, such that $\exists i(F(x^i)) \& \forall y \forall j (F(y^j) \rightarrow y^j \sqsubseteq x^j)$.

A complete compositional semantics for plural definites shows how these denotation conditions are derived from the meaning of ‘the’. We have given such a semantics elsewhere (Nicolas & Payton 2025a: 13–14), but nothing here turns on its details; the two features we rely on can be stated directly. First, this semantics allows ‘the teams’ and ‘the players’ to be co-denoting: a team just *is* the players which make it up, and so ‘the teams’ denotes all the individual players taken together; *they* are the things that can be carved into each of the teams. Second, as explained in Section III.3, a plural definite inherits the basic cover of its head noun. So, $\llbracket \text{the teams} \rrbracket$ inherits from ‘team’ the basic cover $\star^T$; this is a cover defined only for pluralities which can be carved into teams, and which carves any such plurality into all and only the teams included in it.

The semantic value of the preposition ‘of’ is more straightforward; in effect, it stands for neutral inclusion. However, we stipulate that when its first entity argument-place, ‘ x ’, is saturated, the resulting partitive phrase inherits the basic cover of the noun used to saturate that argument-place.

We can now compose $\llbracket \text{of the teams} \rrbracket$. For any two covers, i and j , we obtain the extension, $\llbracket \text{of the teams} \rrbracket^{i,j}$, as follows:

$$\begin{aligned}
 & \llbracket \text{of the teams} \rrbracket^{i,j} \\
 &= \llbracket \text{of} \rrbracket^i(\llbracket \text{the teams} \rrbracket^j) && \text{By FA} \\
 &= \lambda x \lambda y [y^i \sqsubseteq x^i](t) \\
 &= \lambda y [y^i \sqsubseteq t^i]
 \end{aligned}$$

Notice that the cover j has been ‘swallowed up’ in the derivation. It is, therefore, inert: for any i and j , $\llbracket \text{of the teams} \rrbracket^{i,j} = \llbracket \text{of the teams} \rrbracket^i$. So, by abstracting just on i , we recover the intension:

$$\llbracket \text{of the teams} \rrbracket = \lambda i \lambda x [x^i \sqsubseteq t^i]$$

This is the property of being neutrally included in the teams—condition (i). Because ‘ $\sqsubseteq$ ’ is cover-insensitive, the cover variable ‘ i ’ is inert: whether some things satisfy this property depends only on what ‘the teams’ denotes, not on how those things are carved up. Nonetheless, this intension inherits from $\llbracket \text{the teams} \rrbracket^j$ the basic cover $\star^T$, which is what condition (ii) will exploit. It now serves as the argument of $\llbracket \text{one} \rrbracket^i$.

Turning now to $\llbracket \text{one} \rrbracket^i$, we need a semantics of numerals. In the schema below, ‘ $\exists_n$ ’ is an ordinary numerically-specific quantifier (‘there are exactly n things such that...’);²¹ ‘ $\star^\phi$ ’ is the basic cover contributed by the argument ϕ (here,

²¹Our approach to numerically specific quantifiers is the standard one, except with neutral variables in place of the typical singular ones:

$$\begin{aligned}
 \exists_n x(Fx^i) &\equiv_{\text{df}} \exists x_1 \dots \exists x_n \left(Fx_1^i \ \& \ \dots \ \& \ Fx_n^i \ \& \ x_k^i \neq x_m^i \text{ for all } k < m \leq n \right. \\
 &\quad \left. \& \ \forall y \left(Fy^i \rightarrow (y^i = x_1^i \vee \dots \vee y^i = x_n^i) \right) \right)
 \end{aligned}$$

‘of the teams’); and ‘ $\lesssim$ ’ is c-inclusion, so ‘ $y^{\star^{\phi}} \lesssim x^{\star^{\phi}}$ ’, says that y is one of the cells into which $\star^{\phi}$ carves x . The numeral thus requires x to be carved by $\star^{\phi}$ into exactly n cells, and requires x , so carved, to satisfy ϕ . Elsewhere (Nicolas & Payton 2025a: 19–22), we defend this semantics in detail; here we simply state it:

Numeral schema $\llbracket n \rrbracket = \lambda i \lambda \phi \lambda x [\exists_n y (y^{\star^{\phi}} \lesssim x^{\star^{\phi}}) \& \phi(\star^{\phi})(x)]$

Instantiating at $n = 1$ gives the semantic value for ‘one’:

$$\llbracket \text{one} \rrbracket = \lambda i \lambda \phi \lambda x [\exists_1 y (y^{\star^{\phi}} \lesssim x^{\star^{\phi}}) \& \phi(\star^{\phi})(x)]$$

We can now give the full derivation of $\llbracket \text{one of the teams} \rrbracket^i$:

$$\begin{aligned} \llbracket \text{one of the teams} \rrbracket^i &= \llbracket \text{one} \rrbracket^i (\llbracket \text{of the teams} \rrbracket) && \text{By IFA} \\ &= \lambda \phi \lambda x [\exists_1 y (y^{\star^{\phi}} \lesssim x^{\star^{\phi}}) \& \phi(\star^{\phi})(x)] (\lambda j \lambda z [z^j \sqsubseteq t^j]) \\ &= \lambda x [\exists_1 y (y^{\star^T} \lesssim x^{\star^T}) \& \lambda j \lambda z [z^j \sqsubseteq t^j] (\star^T)(x)] \\ &= \lambda x [\exists_1 y (y^{\star^T} \lesssim x^{\star^T}) \& x^{\star^T} \sqsubseteq t^{\star^T}] \end{aligned}$$

This is the property of being included in the teams and being carved into exactly one subplurality by the basic cover $\star^T$. Equivalently: it’s the property of being included in the teams and being a team.

Substituting this into the sentence-level result obtained earlier yields the truth-conditions of (2). Simplifying slightly:

$$\begin{aligned} \llbracket \text{One of the teams is Québécois} \rrbracket^{i,j} &= \llbracket \exists \exists \rrbracket (\llbracket \text{one of the teams} \rrbracket^i) (\llbracket (\text{is}) \text{ Québécois} \rrbracket^j) && \text{By FA} \\ &= \lambda F \lambda G [\exists x (F x \& G x)] (\llbracket \text{one of the teams} \rrbracket^i) (\llbracket (\text{is}) \text{ Québécois} \rrbracket^j) \\ &= \exists x (\llbracket \text{one of the teams} \rrbracket^i(x) \& \llbracket (\text{is}) \text{ Québécois} \rrbracket^j(x)) \\ &= \exists x (\lambda z [\exists_1 y (y^{\star^T} \lesssim z^{\star^T}) \& z^{\star^T} \sqsubseteq t^{\star^T}](x) \& \lambda v [\text{QUÉBÉCOIS}(v^j)](x)) \\ &= \exists x (\exists_1 y (y^{\star^T} \lesssim x^{\star^T}) \& x^{\star^T} \sqsubseteq t^{\star^T} \& \text{QUÉBÉCOIS}(x^j)) \end{aligned}$$

That is, (2) is true just in case some x is a team, is included in the teams, and is Québécois. In our imagined scenario, both teams are from Ontario; so, no team is Québécois and the sentence is false, as desired.

The semantics of (1) is computed in exactly the same way, with ‘players’ replacing ‘teams’ throughout and the constant ‘ p ’ (for the players) replacing ‘ t ’. Letting $\star^P$ be the basic cover of ‘player’, we obtain

$$\begin{aligned} \llbracket \text{One of the players is Québécois} \rrbracket^{i,j} &= \exists x (\exists_1 y (y^{\star^P} \lesssim x^{\star^P}) \& x^{\star^P} \sqsubseteq p^{\star^P} \& \text{QUÉBÉCOIS}(x^j)) \end{aligned}$$

That is, (1) is true just in case, for some x , (i) x is a player, (ii) x is included in the players, and (iii) x is Québécois. In our imagined scenario, the sentence is true, as desired. And this is so despite the fact that (2) is false and ‘the teams’ and ‘the players’ are co-denoting ($p = t$).

On our account both (1) and (2) contain a free cover variable ‘ j ’, relative to which ‘QUÉBÉCOIS(x^j)’ is evaluated. How is j selected? We’ve claimed that the cover counting the satisfiers of ‘one’ is fixed *lexically*: it’s the basic cover of the head noun, inherited from ‘player’ ($\star^P$) or ‘team’ ($\star^T$), and hence settled by the sentence. The variable j is, by contrast, supplied *contextually*, cover-selection being the familiar pragmatic matter of resolving to a salient carving.²² However, while selection of j is largely a pragmatic matter, it’s also *semantically constrained*. Recall that in a quantified sentence the bound variable is restricted by the restrictor nominal and inherits the basic and fitting covers associated with it. As it happens, in (1) and (2) this is idle—‘is Québécois’ is plausibly cover-insensitive in these sentences, and so it pays the basic and fitting covers no mind. In other sentences—e.g. those involving distributive readings—this is important, as it constrains which choices of cover, and which readings of the sentence, are available (footnote 28).

Similar issues arise for choices of predicate other than ‘is Québécois’, but we won’t deal with them here. Our main aim has been to show that (1) and (2) can have different truth-values *even if ‘the players’ and ‘the teams’ are co-denoting*. Key to this result is our semantics of partitives. The original regimentation of the argument from Section 1 assumed that the English ‘is one of’ is correctly regimented as ‘ $\leq$ ’, which in turn is understood so that ‘ $x \leq y$ ’ is transparent: whether x is *singularly* included in y depends entirely on what ‘ x ’ and ‘ y ’ denote. On our view, this is incorrect. Although ‘the players’ and ‘the teams’ denote the same plurality, the English partitives ‘one of the players’ and ‘one of the teams’ don’t both denote the property of being an individual in that plurality. Instead, each phrase denotes the property of being included in that plurality *and* being carved into exactly one subplurality by the relevant basic cover, where this cover is contributed by the head noun— $\star^P$ by ‘player’, $\star^T$ by ‘team’. As a result, whether some thing(s) count as ‘one of the players’ or ‘one of the teams’ depends, not just on what ‘the players’ or ‘the teams’ denotes, but on how that denotation has been carved up.

It’s important to note that this isn’t an *ad hoc* solution. The use of covers in the semantics of plural predication is independently motivated, since it lets us treat a variety of plural predicates—‘overlap’ among them—without invoking controversial higher-level plural resources. The same is true for our semantic framework, on which count nouns are associated with basic and fitting covers, and numerals are forced to be interpreted with respect to basic covers. So,

²²On the pragmatics of cover-selection see e.g. Gillon (1990: 482–3), Schwarzschild (1996: 92–8), and Grimau (2021b: 179).

our approach to partitives is motivated independently of the identity theory of groups.

Moreover, recall that it's commonly accepted that at least *some* collective nouns are pseudo-singular. That view faces a problem perfectly analogous to the one confronting the identity theory of groups. Consider:

(8) One of the socks is on sale.

(9) One of the pairs of socks is on sale.

Imagine you're shopping and have several pairs of socks in your cart. Most are sold at regular price, but one pair is half off; the store sells socks only in pairs. Here (8) seems false while (9) seems true. Yet if we regiment them as the plural logician would, then on the assumption that 'the socks' and 'the pairs of socks' co-denote, (8) and (9) come out equivalent. Similar problems can be raised for other collective nouns. So, if you think that *any* collective nouns are pseudo-singular, you have good reason to accept our semantics for partitives.

V. Composition as identity

As announced in Section I, the identity theory of groups is a special case of a more general idea: that a complex object just *is* the things that make it up. Its best-known instance concerns not groups and their members but wholes and their parts. According to *composition as identity* (CAI), a whole is identical to its parts—identical to *all* of them collectively, not to *each* of them individually.²³ For instance, a cat is identical to all its parts—its head, torso, legs and tail taken together—but not to any one of them. CAI faces an objection exactly parallel to the one we raised against the identity theory in Section I, and turning on the very same partitive construction. The diagnosis and the cure carry over with almost no change.

The objection originates with Yi (1999: 146). Suppose Tom and Jerry compose an object, 'Genie'.²⁴ Genie is one of Genie; so, by Leibniz's Law, if CAI is true then Genie is one of Tom and Jerry. But Genie is *not* one of Tom and Jerry; only Tom and Jerry are. How can that be, if Genie just *is* Tom and Jerry? As it stands the argument is ungrammatical—neither 'one of Genie' nor 'one of Tom and Jerry' is acceptable, since the embedded noun phrase can be neither a proper name nor a conjunction of proper names²⁵—but it's readily

²³See notably Wallace (2011a, b), Bohn (2014), and Payton (2021, 2022). The opposing view treats a whole as an individual distinct from its parts; see e.g. Sider (2007, 2014), Yi (2014), and Calosi (2016).

²⁴Yi defines composition thus: a plurality *y* composes an individual *x* iff every individual *u* among *y* is part of *x* and every part of *x* overlaps some individual *v* among *y*. Equivalently, *x* is a mereological sum of *y*; this is one standard definition of that notion (Cotnoir & Varzi 2021).

²⁵See Roberts (1987: 150), Reed (1996: 151), and de Hoop (1997: 161). Yi (1999: 147) circumvents the difficulty by a similar manoeuvre to the one we adopt.

reconstructed with grammatical materials. Let Butch be distinct from Tom and Jerry, and consider:

(10) Genie is one of them.

Read ‘them’ as anaphoric on ‘Butch, Tom, and Jerry’: since Genie isn’t one of *them*, (10) is false. Now read ‘them’ as anaphoric on ‘Butch and Genie’: since Genie is one of *them*, (10) is true. Yet if CAI is true, Butch and Genie just *are* Butch, Tom, and Jerry (Genie being identical to Tom and Jerry), so ‘them’ co-denotes on the two readings and (10) should receive a single truth-value. It doesn’t.²⁶

As before, the argument is bolstered by regimentation in plural logic. Letting the constant ‘*t*’ go proxy for the pronoun ‘them’, the plural logician renders (10) as ‘ $g \leq t$ ’. Because ‘ $\leq$ ’ is transparent, whether $g \leq t$ turns only on what ‘*g*’ and ‘*t*’ denote; so if ‘them’ co-denotes across the two readings, the regimented sentence cannot change truth-value, and CAI is apparently refuted.

The diagnosis is the one given in Sections II and IV. The transparent ‘ $\leq$ ’ is the wrong regimentation of ‘is one of’. As with ‘one of the *F*s’, the predicate ‘is one of them’ makes two demands on its satisfiers: (i) *inclusion*, contributed by ‘of’, which is cover-insensitive and depends only on what ‘them’ denotes; and (ii) a *count relative to a carving*, contributed by ‘one’, which requires the satisfiers to constitute exactly one cell of a basic cover. The only structural novelty is the source of that cover. With ‘one of the teams’ the cover was inherited from the head noun. Here there is no head noun: ‘them’ inherits the basic cover of its antecedent. And co-denoting antecedents can contribute different covers—‘Butch, Tom, and Jerry’ carves the common denotation into Butch, Tom, and Jerry, whereas ‘Butch and Genie’ carves it into Butch and Genie—so that ‘is one of them’ expresses different properties on the two readings even though ‘them’ co-denotes. Genie constitutes a cell of the second carving but not of the first, and so (10) comes out true on the one reading and false on the other, exactly as it should, with no threat to CAI.

We now make this precise. The partitive here occupies predicate rather than subject position, which in fact simplifies the derivation: no unpronounced existential operator is needed. At its most abstract

$$\begin{aligned} & \llbracket \text{Genie is one of them} \rrbracket^{i,j} \\ &= \llbracket (\text{is}) \text{ one of them} \rrbracket^i (\llbracket \text{Genie} \rrbracket^j) \end{aligned} \quad \text{By FA}$$

Letting *g* be the plurality denoted by ‘Genie’, we give that name the semantic value

$$\llbracket \text{Genie} \rrbracket = \lambda i[g]$$

²⁶McKay (2016: 178) develops a parallel argument against identifying the denotations of mass nouns with pluralities (as Nicolas (2008) does). Our response extrapolates straightforwardly.

The meaning of $\llbracket(\text{is}) \text{ one of them} \rrbracket^i$ requires more comment. In (10), ‘them’ is anaphorically related to a given noun phrase—in our imagined cases, ‘Butch, Tom, and Jerry’ or ‘Butch and Genie’. By the hypothesis of CAI, these noun phrases denote the same plurality t ,²⁷ and so we can make use of the following entry for ‘them’:

$$\llbracket\text{them}\rrbracket = \lambda i[t]$$

The intension, $\llbracket\text{of them}\rrbracket$, is then derived perfectly analogously to $\llbracket\text{of the teams}\rrbracket$:

$$\llbracket\text{of them}\rrbracket = \lambda i \lambda x[x^i \subseteq t^i]$$

The extension, $\llbracket(\text{is}) \text{ one of them} \rrbracket^i$, is explicated below. As previously, the expression inherits the basic cover, $\star$, associated with the embedded noun phrase, in this case the pronoun ‘them’.

$$\begin{aligned} & \llbracket(\text{is}) \text{ one of them} \rrbracket^i \\ &= \llbracket\text{one}\rrbracket^i(\llbracket\text{of them}\rrbracket) && \text{By IFA} \\ &= \lambda \phi \lambda x[\exists_1 y(y^{\star\phi} \lesssim x^{\star\phi}) \ \& \ \phi(\star^\phi)(x)](\lambda k \lambda z[z^k \subseteq t^k]) \\ &= \lambda x[\exists_1 y(y^{\star} \lesssim x^{\star}) \ \& \ \lambda k \lambda z[z^k \subseteq t^k](\star)(x)] \\ &= \lambda x[\exists_1 y(y^{\star} \lesssim x^{\star}) \ \& \ x^{\star} \subseteq t^{\star}] \end{aligned}$$

So, the complete derivation of (10) is

$$\begin{aligned} & \llbracket\text{Genie is one of them}\rrbracket^{i,j} \\ &= \llbracket(\text{is}) \text{ one of them}\rrbracket^i(\llbracket\text{Genie}\rrbracket^j) && \text{By FA} \\ &= \lambda x[\exists_1 y(y^{\star} \lesssim x^{\star}) \ \& \ x^{\star} \subseteq t^{\star}](g) \\ &= \exists_1 y(y^{\star} \lesssim g^{\star}) \ \& \ g^{\star} \subseteq t^{\star} \end{aligned}$$

That is, g are included in t and $\star(g)$ has a single cell.

So far, we’ve left the basic cover as the schematic $\star$. As anticipated in the sketch above, the pronoun inherits the basic cover of its antecedent; it remains to say what the basic covers of ‘Butch, Tom, and Jerry’ and ‘Butch and Genie’ are.

²⁷Although a semantics for term-forming ‘and’ isn’t strictly necessary for our purposes, one is easily provided:

$$\llbracket\text{and}\rrbracket = \lambda i \lambda x \lambda y[\lambda z(\forall v(Iv^i \rightarrow (v^i \subseteq x^i \leftrightarrow (v^i \subseteq y^i \vee v^i \subseteq y^i))))]$$

where ‘I’ is the individuality predicate of Section 1 and ‘ ι ’ the definite description operator. That is, $\llbracket\text{and}\rrbracket$ takes two pluralities to the plurality of all and only the individuals among one or the other of them. So, for instance, $\llbracket\text{Tom and Jerry}\rrbracket^i$ denotes the plurality of all and only the individuals among Tom or among Jerry. On the assumption—unfriendly to CAI—that Tom and Jerry are *individuals*, including nothing other than themselves, this plurality includes Tom, Jerry, and nothing else. But if CAI is true, it includes many other things, i.e. all of their proper parts.

We take it that the basic cover of a proper name simply maps its denotation to itself: $\star^B$ maps the denotation of ‘Butch’ to Butch, $\star^T$ maps that of ‘Tom’ to Tom, etc. And we take it that the basic cover of a compound term is derived from the basic covers of its components as follows:

Basic covers (Compounds) Where α is a compound term with constituent terms $\alpha_1, \dots, \alpha_n$, the basic cover $\star^\alpha$ is defined for some x whenever the values that $\star^{\alpha_1}, \dots, \star^{\alpha_n}$ assign to their arguments and which are included in x leave nothing among x out; its values at x are all and only those.

This is plausible independently of its application to compound *names*. Consider:

(11) The choirs and the string quartets met.

There’s a reading of this sentence on which ‘met’ distributes over each of the choirs and over each of the string quartets: on that reading, (11) is true just in case (i) each of the choirs held a meeting and (ii) each of the string quartets held a meeting. Letting c be the choirs, s the string quartets, and cs the choirs and string quartets taken together: $\star^C$ carves c into all and only the choirs among them; $\star^S$ carves s into all and only the string quartets among them; and so by our principle, $\star^{CS}$ carves cs into all and only the things among them that are either choirs or string quartets. On our view, distributive predicates must be interpreted with respect to covers which fit the noun. The ‘pure distributive’ reading is obtained when the *basic* cover is chosen (Nicolas & Payton 2025a: 16). Here, this is $\star^{CS}$. So, ‘met’ distributes over each member of $\star^{CS}$, i.e. over each choir and each string quartet.

So, coming back to our truth-conditions for (10),

$$\exists_1 y(y^* \lesssim g^*) \ \& \ g^* \sqsubseteq t^*$$

the schematic $\star$ is resolved by the antecedent, and the two readings come apart exactly as anticipated. When ‘them’ is anaphoric on ‘Butch, Tom, and Jerry’, $\star = \star^{BTJ}$, which carves t into Butch, Tom, and Jerry; and since Tom and Jerry are both included in g while Butch is not, $\star^{BTJ}$ carves g into *two* cells, Tom and Jerry. So the first conjunct fails and (10) is false—even though $g \sqsubseteq t$.

When ‘them’ is anaphoric on ‘Butch and Genie’, $\star = \star^{BG}$; here Genie alone is included in g , so $\star^{BG}$ carves g into a single cell, and— $g \sqsubseteq t$ holding as before—(10) is true. So Genie can be one of Butch and Genie while failing to be one of Butch, Tom, and Jerry, even though, by CAI, these co-denote. The inclusion condition (i) is satisfied on both readings; it’s condition (ii), the count relative to the antecedent’s basic cover, that distinguishes them.

VI. Conclusion

A simple semantics for plural partitives—expressions of the form [Det [of NP]], where the embedded noun phrase is plural—takes the preposition ‘of’ to indicate inclusion while the determiner adds further information, e.g. that the satisfiers are n in number. We’ve argued that this semantics, while initially plausible, is false. While ‘of’ indicates inclusion—in the sense of neutral inclusion, or $\sqsubseteq$ —its interaction with determiners renders plural partitives cover-sensitive, in a way not typically recognized.

We began with a problem for the identity theory of groups (Section I): a simple argument involving a plural partitive seemed to show that the teams cannot be the players. We then found that composition as identity faces a structurally identical problem (Section V): the same partitive construction seemed to show that a whole cannot be its parts. Both arguments share a single defective assumption about how ‘is one of’ should be regimented—an assumption our semantics rejects. Replacing it (Sections III–V) dissolves both objections at once: neither the argument against the identity theory of groups (Section IV) nor the argument against composition as identity (Section V) survives. One semantics for plural partitives thus answers two independently motivated objections to two metaphysical views, and this is good news for anyone who endorses either or both.

It’s worth being explicit about the strength of this result, since a critic might object that our semantics simply builds in the metaphysics under dispute, so that its availability is no evidence for that metaphysics. We grant the conditional point: the bare existence of a friendly semantics does not, by itself, establish composition as identity or the identity theory of groups. Our aim is more modest and defensive. The partitive objections, if sound, would be decisive; we show that they aren’t sound, because they rest on a contestable regimentation of ‘is one of’ as the cover-insensitive ‘ $\leq$ ’. That regimentation isn’t neutral ground that the identity theorist must accept; it’s itself a substantive semantic commitment. Removing a putatively devastating objection isn’t the same as proving the view, but it’s a substantive result.

Moreover, the semantics we deploy isn’t adopted *ad hoc* to rescue these views: covers are motivated independently, by distributivity (4) and by the avoidance of superplurals (3), and the cover-sensitivity of ‘is one of’ is forced by considerations—notably pseudo-singular collective nouns like ‘pair’ and ‘trio’—that arise whether or not one holds either metaphysical view. Indeed, since it’s widely accepted that at least some collective nouns are syntactically singular yet semantically plural (e.g. ‘pair’, ‘trio’), virtually everyone needs a non-standard partitive semantics (cover-sensitive or higher-level plural) for those nouns. Extending the account to all collective nouns is then *attractive*, since it yields a unified theory on which collective nouns carry no special ontological commitments.

Our approach involves a significant departure from standard plural logic. Recall that a typical plural logician glosses ‘is one of’ as singular inclusion. So, she regiments any sentence of the form ‘ a is one of b ’ as ‘ $a \leq b$ ’, where ‘ $\leq$ ’ is cover-insensitive, and indeed, covers make no appearance in the formula *at all*. This is what generates the problem for CAI. We think that the logical form of such sentences is better represented, in CPL, as

$$\exists_1 x(x^* \lesssim a^*) \ \& \ a^* \sqsubseteq b^*$$

where $\star$ is the basic cover for the term ‘ b ’. Likewise, the plural logician regiments any sentence of the form ‘One of the F s is G ’ as ‘ $\exists x(x \leq f \ \& \ Gx)$ ’, where the cover-insensitivity of ‘ $\leq$ ’ once again creates a problem, this time for the view that groups are identical to their members. We think that the logical form of such sentences is better represented, in CPL, as

$$\exists x(\exists_1 y(y^* \lesssim x^*) \ \& \ x^* \sqsubseteq f^* \ \& \ Gx^j)$$

where $\star$ is the basic cover for ‘the F s’.

However, we don’t abandon singular inclusion; we subsume it. Since ‘ $a \leq b$ ’ just is ‘ $Ia \ \& \ a \sqsubseteq b$ ’, cover-insensitivity is recovered as a limit case: whenever the basic cover $\star$ atomizes the relevant plurality—carving it into its individual members, as it does paradigmatically when the head noun or antecedent is true of individuals—the first conjunct $\exists_1 x(x^* \lesssim a^*)$ reduces to Ia , and our truth-conditions collapse into ‘ $a \leq b$ ’. The standard regimentation is thus the special case of ours in which carving makes no difference. What we add is the recognition that carving sometimes *does* make a difference, and that natural-language partitives track it.

The arguments we’ve considered used partitives with the numeral ‘one’. But given our general schema for numerals (Section IV), similar remarks can be made for any numerically specific partitive. For instance, suppose again that there are two teams from Ontario: Ann and Bill, and Carrie and Daniel. But suppose this time that both Ann and Daniel are Québécois: each team has exactly one Québécois member. Then, (12) is true while (13) is false.

(12) Two of the players are Québécois.

(13) Two of the teams are Québécois.

This might appear to be impossible if the teams are identical to the players, but our account shows it to be possible. The derivations proceed similarly to those for (1) and (2), with $\exists_2$ in place of $\exists_1$.²⁸

Indeed, we’ve laid the foundation for a general approach to plural partitives. Here, we’ve focused on numerically specific ones. But analogous

²⁸Here ‘are Québécois’ is *distributive*, so the truth-conditions of the two sentences are

$\llbracket \text{Two of the players are Québécois} \rrbracket^{i,j}$

$= \exists x(\exists_2 y(y^{*p} \lesssim x^{*p}) \ \& \ x^{*p} \sqsubseteq p^{*p} \ \& \ \text{QUÉBÉCOIS}^D(x^j))$

treatments of numerically unspecific ones suggest themselves. We might, for instance, give the determiners ‘some’ and ‘all’ the following semantic values:

$$\llbracket \text{some} \rrbracket = \lambda i \lambda \phi \lambda x [\exists y (y^{*\phi} \lesssim x^{*\phi}) \ \& \ \phi(\star^\phi)(x)]$$

$$\llbracket \text{all} \rrbracket = \lambda i \lambda \phi \lambda x [\forall y (\exists z (\phi(\star^\phi)(z) \ \& \ y^{*\phi} \lesssim z^{*\phi}) \rightarrow y^{*\phi} \lesssim x^{*\phi}) \ \& \ \phi(\star^\phi)(x)]$$

That ‘some’, ‘all’, and each numeral fall under a single functional type highlights a further advantage of our semantics over alternatives: its *compositional* account of the meanings of partitives. The plural logician, by contrast, takes ‘is one of’ as a primitive true of an individual and a plurality, and reaches ‘two of’, ‘some of’, and ‘all of’ only by paraphrase into plural logic—expressible, but not compositional. Much the same is true of the higher-level pluralist, with an added wrinkle. Many systems of higher-level plural logic make use of *level-specific* inclusion relations: $\leq^1$ relates individuals to ordinary, ‘first-level’ pluralities; $\leq^2$ relates ordinary pluralities to ‘second-level’ pluralities, etc.²⁹ If she goes this way, then the higher-level pluralist distinguishes (1) and (2)—or similar sentence-pairs—by claiming that ‘the teams’ denotes a higher-level plurality, so that ‘is one of’ means $\leq^1$ in (1) but $\leq^2$ in (2). So, like the plural logician, she lacks a compositional story of the meanings of partitives, and she may even need to treat them all as type-ambiguous. We incur neither cost.

A complete semantics for plural partitives lies beyond this paper. But we have laid the foundation for a general and attractive account, with significant consequences for debates in metaphysics.³⁰

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$\llbracket \text{Two of the teams are Québécois} \rrbracket^{i,j}$

$$= \exists x (\exists_2 y (y^{*T} \lesssim x^{*T}) \ \& \ x^{*T} \sqsubseteq t^{*T} \ \& \ \text{QUÉBÉCOIS}^D(x^j))$$

A distributive predicate is evaluated at a fitting cover of its argument, and so never distributes below the cells fixed by the noun. Since ‘*x*’ inherits the fitting covers of ‘player’ in (12) and of ‘team’ in (13), admissible values of *j* confine distribution in the former to individual players and arbitrary combinations thereof, and in the latter to teams and arbitrary combinations thereof—but *never* to individual players. On the pure distributive reading, where *j* is the basic cover, (13) is accordingly true iff some *x* is carved into two teams each of which is Québécois, i.e. each affiliated *qua* team with a city in Québec.

²⁹See especially Rayo (2006). See Oliver & Smiley (2016) for an alternative.

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